

Recent Progress in Asymmetric Allylic Substitutions Catalyzed by Chiral Copper Complexes

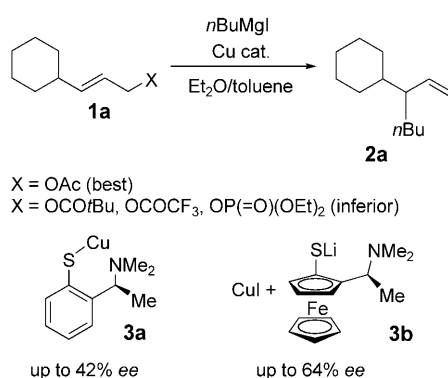
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Allylic substitutions (also known as allylic alkylations) are among the most important carbon–carbon bond-forming reactions in organic synthesis. Palladium-catalyzed allylic substitutions and their asymmetric version have been extensively studied and widely used in a variety of total syntheses.^[1] Palladium catalysis mostly requires soft nucleophiles such as malonate carbanions for high stereo- and regioselectivity. On the other hand, copper catalysis allows the use of hard nucleophiles including Grignard and organozinc reagents, thus being complementary to palladium catalysis. Moreover, copper-catalyzed substitution reactions usually proceed with high S_N2' regioselectivity, which guarantees the creation of stereogenic centers. Despite its promising potential, copper-catalyzed allylic substitution reactions had fallen behind, especially with regards to the development of asymmetric reactions.^[2] However, since the first report on asymmetric allylic substitutions catalyzed by a chiral copper complex appeared in 1995,^[3] remarkable progress in the development of copper-catalyzed asymmetric reactions has been made.

Bäckvall, van Koten, and co-workers reported that chiral arenethiolato-copper(I) **3a** catalyzed the asymmetric allylic substitution of allylic acetate with

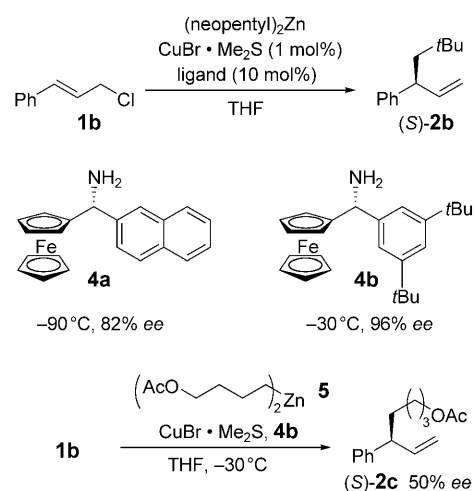


Scheme 1.

butylmagnesium reagent (Scheme 1).^[3] Their contribution is quite important not only because it reported the first copper-catalyzed asymmetric allylic substitution but also as the inherent difficulty of this asymmetric transformation was pointed out. The enantiomeric excess of the product heavily depended on many factors including the coordinating ability of the leaving group in the allylic substrates, the temperature, and the method of addition of the substrate and the Grignard reagent. The reactions of butylmagnesium iodide with **1a** (X = OAc) catalyzed by **3a** at 0°C and at 20°C showed similar enantioselectivities (29% *ee* and 31% *ee*, respectively). A lower temperature of –20°C adversely affected the enantioselectivity (7% *ee*). Simultaneous addition of the substrate and the Grignard reagent to a solution of **3a** provided the product with 42% *ee*. Their observations underscored the difficulty in generating a unique transition state that leads to high

enantioselectivity; there appeared to be several species present in the reaction mixture. Bäckvall and co-workers recently obtained the highest *ee* value (64% *ee*) for the reaction of **1a** with butylmagnesium iodide in the presence of ferrocenethiolato-copper(I) **3b**.^[4] Even with this latest ligand **3b**, fine tuning of the solvent, temperature, and method of addition was required.

A breakthrough in copper-catalyzed asymmetric allylic substitutions came in 1999,^[5] when Dübner and Knochel focused on dialkylzinc as an alkyl source (Scheme 2). To attain a high enantioselectivity, the system required a high ratio of ligand to copper, very low temperatures, and the presence of bulky alkyl groups on zinc. For instance, the reaction of cinnamyl chloride with di-*n*-pentylzinc under catalysis by CuBr·Me₂S/**4a** (1:10) at –90°C afforded (*S*)-**2b** with 82% *ee*. Note, no enantioselectivity was observed when alkylzinc chloride, RZnCl, was used instead of



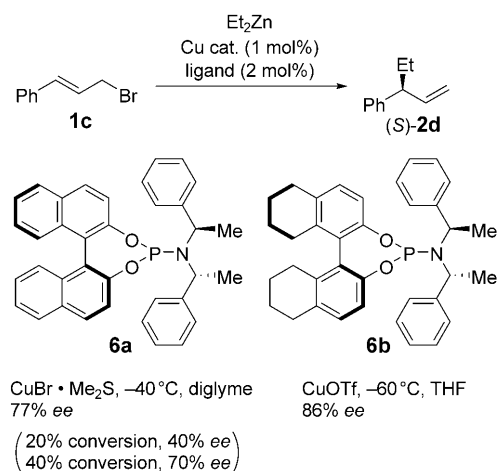
Scheme 2.

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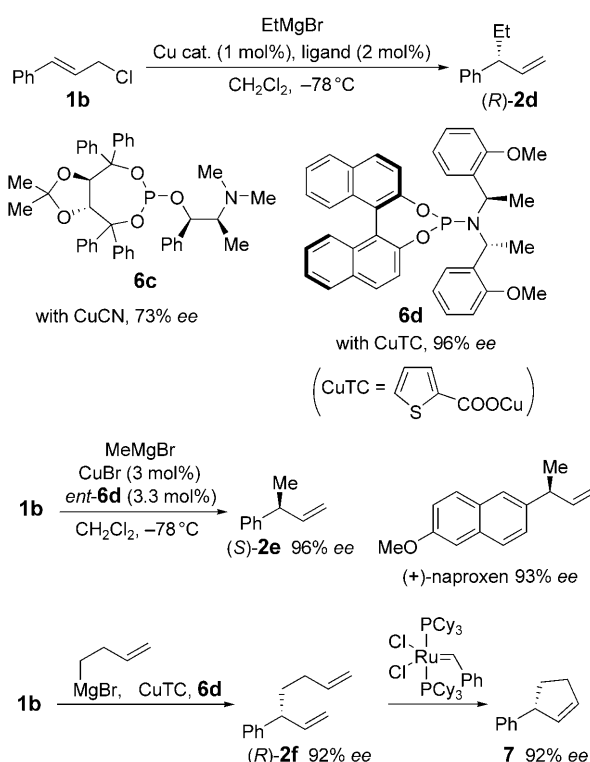
dialkylzinc, R_2Zn (see Woodward's work^[6]). Later, ligand **4b** proved to be more effective in this reaction to provide **2b** with 96% *ee*. However, dineopentylzinc still gave the best results. The use of linear dialkylzincs such as dipentylzinc and functionalized dialkylzinc **5** afforded the corresponding products with approximately 44–65% *ee*. Despite the necessity of bulky alkyl groups, the work of Dübner and Knochel contributed to the accelerated development of copper-catalyzed asymmetric allylic substitution reactions.

Highly enantioselective allylic substitutions with linear alkylzinc reagents remained a major challenge, but then Feringa and co-workers proposed chiral phosphoramidite ligands as the solution (Scheme 3).^[7] Treatment of cinnamyl bromide with diethylzinc in the presence of $CuBr \cdot Me_2S$ coupled with phosphoramidite **6a** in diglyme at $-40^\circ C$ afforded **2d** with 77% *ee*. Importantly, the enantiomeric excess increased as the reaction proceeded, and it was rationalized that different catalytic species are present at the early stage of the reaction. Further extensive screening of chiral phosphoramidites revealed that **6b** led to improved enantioselectivities.

Independently, Alexakis and co-workers developed a system similar to that reported by Feringa for the reaction of Grignard reagents (Scheme 4).^[8] As Bäckvall and van Koten also noted, enantioselectivity as well as regioselectivity were functions of many variables, even with the aid of chiral phosphite ligand **6c**. Very recently, a highly effective



Scheme 3. Diglyme = diethyleneglycol dimethyl ether, Tf = trifluoromethanesulfonyl.



Scheme 4. Cy = cyclohexyl.

phosphoramidite ligand **6d** was developed that provided **(R)-2d** with high enantioselectivity. This ligand was also suitable for asymmetric allylic substitutions with diethylzinc (91% *ee*, starting from cinnamyl bromide, $-40^\circ C$) and with methylmagnesium bromide (96% *ee*). The highly enantioselective methylation was applied to the formal synthesis of (+)-naproxen. Asymmetric substitution with 3-butenyl Grignard reagent followed by ring-closing metathesis in one pot yielded chiral cyclopentene **7**, a new chiral synthon (Scheme 4). At present, this is the most efficient asymmetric allylic substitution reported with a Grignard reagent.

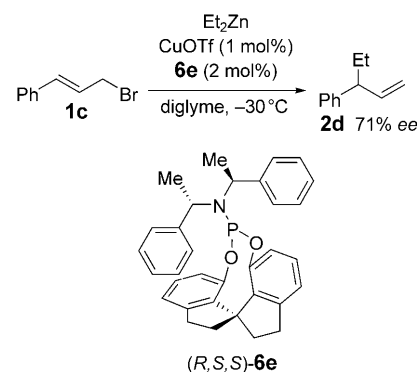
Zhou and co-workers devised original chiral spiro phosphoramidite and phosphite ligands and applied them to asymmetric allylic substitution reactions (Scheme 5).^[9] Enantiomeric excess of the products was generally moderate. They successfully obtained a single crystal of a copper complex from $CuBr \cdot Me_2S$ and a spiro phosphoramidite ligand, **L**, which was similar to **6e**. X-ray

crystallographic analysis disclosed the dimeric structure of $[CuBrL_2]_2$. Furthermore, the optimal Cu/ligand ratio for the reaction was shown to be 1:2. Although the actual shapes of the active species and the transition state are still not clear, these observations are nonetheless quite informative.

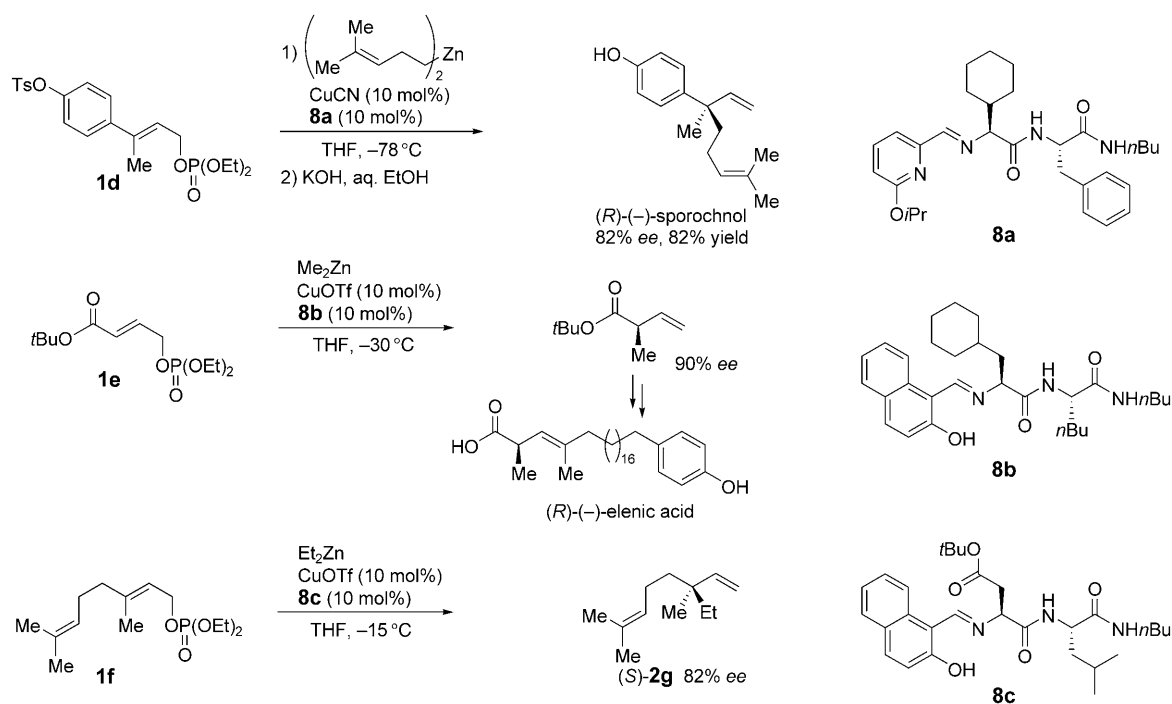
Hoveyda and co-workers developed various peptide-based chiral ligands for asymmetric allylic substitution reactions under copper catalysis (Scheme 6).^[10]

Through the synthesis and analysis of a library of ligands, the modularity of these ligands could be utilized for the improvement of

the efficiency, regioselectivity, and stereoselectivity of the reaction. Dipeptide ligands **8** permitted access to the asymmetric syntheses of quaternary carbon atoms and α -alkyl- β,γ -unsaturated esters starting from allylic phosphates such as **1d–1f** (Scheme 6). The reactions were applied to the syntheses of a natural product, **(R)-(-)-spirochnol**, and the topoisomerase II inhibitor, **(R)-(-)-elenic acid**. With the state-of-the-art ligand **8c**, efficient addition of the alkyl reagent to trisubstituted aliphatic, not aromatic, substrates to deliver quaternary carbon centers (from **1f** to **2g**, for instance; see Scheme 6) was



Scheme 5.



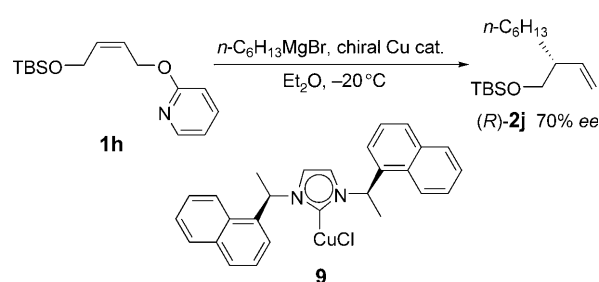
Scheme 6. Ts = *p*-toluenesulfonyl.

observed. Furthermore, the authors proposed a plausible working model through extensive screening of peptide ligands.

Piarulli, Gennari, and co-workers applied their own library of Schiff bases to the asymmetric desymmetrization of *meso* cyclic allylic bis(diethylphosphates) with organozinc reagents (Scheme 7).^[11] Although only the reactions of a 4-cyclopentene-1,3-diol derivative gave satisfactory results, ligands **8d** and **8e** exhibited high efficiencies. Interestingly, in the case of asymmetric

phenylation, selective transfer of the phenyl group from a mixed organozinc reagent ($2\text{Ph}_2\text{Zn} + \text{Me}_2\text{Zn}$) furnished the phenylated product with good enantioselectivity.

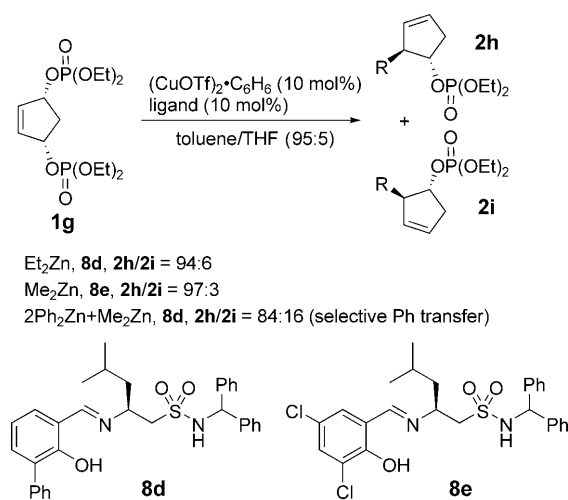
Okamoto and co-workers used C_2 -symmetric carbene-copper complexes for asymmetric allylic substitution reactions (Scheme 8).^[12] Sterically demanding **9** gave the highest *ee* values. The 2-pyridyloxy group of **1h** as well as the acetoxy group were good leaving groups. Although a moderate level of *ee* was obtained in preliminary results, the use of readily modifiable carbene ligands merits further investigation.



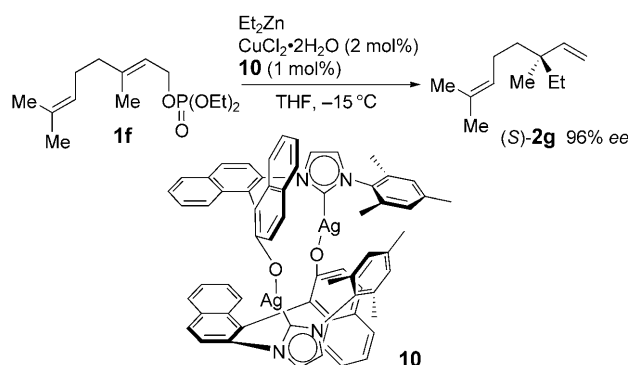
Scheme 8. TBS = *tert*-butyldimethylsilyl.

went facile ligand exchange with copper salts to generate highly effective chiral copper complexes. The catalyst loading was significantly lower than required with Schiff base ligands such as **8c**, and a higher extent of asymmetric induction in the alkylation of trisubstituted olefins than that obtained in Scheme 6 was noted. An advantage of the system is that air- and moisture-insensitive $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ can catalyze the asymmetric reactions.

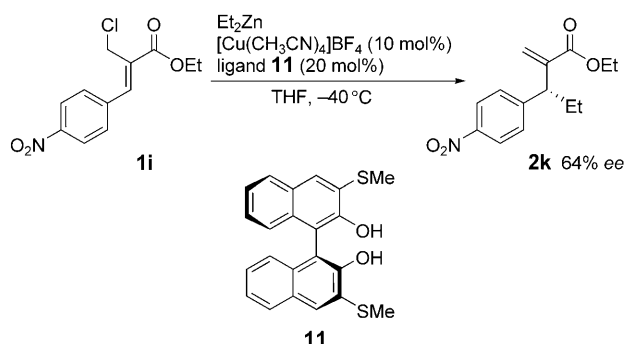
Woodward and co-workers examined the asymmetric allylic substitution of 2-halomethylcinnamate derivatives (Scheme 10),^[14] which are attractive substrates as they are readily prepared on a large scale in two steps through Morita–Baylis–Hillman chemistry. The first generation BINOL (BINOL = 2,2'-binaphthol) catalyst **11** delivered the



Scheme 7. Ph = phenyl.



Scheme 9.



Scheme 10.

corresponding products with up to 64% ee. Other BINOL-type ligands exhibited lower enantioselectivities, and the system resisted further optimization. Very recently, Woodward and co-workers identified chiral secondary amines as alternative ligands for the asymmetric conversion of 2-halomethylcinnamates (Scheme 11).^[6] The C_2 -symmetric di(1-arylethyl)amine skeleton was indispensable for efficient stereoselection. By means of high-throughput screening, 4-methoxy-substituted ligand **12** appeared to be the optimal ligand.

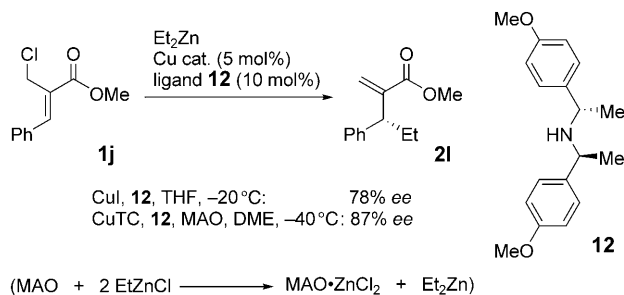
During the high-throughput screening of ligands, Woodward and co-workers

noticed that free amines such as **12** were always superior to their corresponding hydrochloride salts (Scheme 11). Moreover, kinetic studies revealed that the enantioselectivity of the reaction decreased as the reaction proceeded. Also, as observed by Dübner and Knochel,^[5] the deliberate use of ethylzinc chloride instead of diethylzinc also led to poorer enantioselectivity. It was hence expected that removal of EtZnCl , which was gradually formed during the reaction, would enhance the enantioselectivity. To this end, they decided to exploit the zinc Schlenk equilibrium with the aid of methylalu-

minoxane (MAO). Indeed, the addition of MAO resulted in the anticipated increase in enantioselectivity. After further optimization, they attained up to 87% ee (CuTC (copper(I) thiophene-2-carboxylate), **12**, MAO, DME; Scheme 11). Although the difference in the enantiomeric excess in the absence and presence of MAO was not so large, the use of the zinc Schlenk equilibrium is attractive and applicable to other reactions.^[15] The report by Woodward and co-workers lacks detailed investigation on the zinc Schlenk equilibrium that featured in their reaction. Thus, further mechanistic studies and more-efficient systems are awaited.

Copper salts are generally cheaper than those of palladium. More significant advantages are that copper-catalyzed allylic substitutions normally proceed with high $\text{S}_{\text{N}}2'$ regioselectivity and that copper catalysis allows the use of hard nucleophiles. Further extension of the scope of organometallic reagents may lead to considerably complex chiral organic molecules: for instance, the use of arylmetal reagents has not been well exploited. Although an increasing number of copper-catalyzed asymmetric allylic substitutions have emerged, little attention has been paid to their mechanistic aspects. Identification of the active species in such reactions is also still in its infancy. On the basis of experimental and theoretical studies on simpler systems,^[16] profound mechanistic understanding of the enantioselection should open the door to more diverse copper-catalyzed asymmetric allylic substitutions.

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Scheme 11. DME = 1,2-dimethoxyethane.

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